# When the Top Answer Decides: Gender Bias in Decision Models

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## Abstract

Decision models return option probabilities for typed questions in one forward pass, and applications act on the top answer. We test TypeSafe's closed Jev and five open decision models with the gender-bias studies of Fulgu and Capraro (2024a), BBQ, StereoSet and a gender-swapped decision task (Tamkin et al., 2023). All six rate violence for a greater good as more acceptable against a man than against a woman, in every act and stake. Asked who wrote a child's phrase without a gender cue, two open models answer "girl" in 82–85% of cases, but most abstain when they may. In ambiguous BBQ gender questions Jev always abstains; the open models commit in 8–61%, mostly to the stereotyped one. In the decision task, five of the six give women significantly more favourable answers (+0.05 to +0.10 in log-odds); a gender swap changes 1–3.5% of top answers (Jev's partly through call-to-call variation). The top answer turns leans into firm decisions (clef-flash: 0.59 to 0.89), and a confidence gate removes many but mostly not their direction. Two open models described as students of Jev are 3–9 times further from it on our questions than their cards report for their own test set.

## 1 Introduction

Decision models are built to act. In September 2026 TypeSafe AI released Jev, "a new class of frontier models built to make fast, structured decisions that software can use directly" (Almeida, 2026): it answers typed questions (pick one of these options, score this on a scale, yes or no) with option probabilities from one forward pass, through a closed API. Open decision models from Perplexity, Cloudflare, IBM, Liquid AI and Jared Palmer can be served through llama.cpp's `/v1/systemone` endpoint (Perplexity, 2026; Cloudflare, 2026; IBM Granite, 2026; Liquid AI, 2026; Palmer, 2026); the clef-flash and Kev cards state compatibility with Jev's API. AutoTrust's cards describe its open JEV models as students of Jev 1.13 (AutoTrust AI, 2026b,a). An application acts on the top answer, so a bias can become a different decision about a person: an allocational harm, not only a representational one (Gallegos et al., 2024).

The makers add caveats. AutoTrust writes "Not for high-stakes decisions" (AutoTrust AI, 2026b); Kev's model card says "Do not use it for consequential automated decisions about people without human review. Biases of the base model and of the training data ... are not measured" (Palmer, 2026). We found no bias measurement in the cards and announcements we checked.

Fulgu and Capraro (2024a) found that GPT-4 attributed masculine-stereotyped children's phrases to girls far more readily than feminine ones to boys, and judged violence against a man more acceptable than against a woman when a nuclear apocalypse was at stake. We ask whether such biases appear in decision models. Our contributions:

1. A bias evaluation of Jev and five open decision models at their released 16-bit precision (the 27B model in 8-bit): the original paradigms extended by prompt variants, five stakes and an abstention option; BBQ in nine categories; StereoSet; and a gender-swapped decision task that tests the allocation claim directly.

2. A comparison of probability-level measures with the argmax (top-answer) decision, with and without the makers' confidence gate.

3. A comparison of two open models described as students of Jev with their teacher.

4. Controls for base model, interface, numeric precision and serving path.

*This work was done independently and is not endorsed by any lab or by the author's employer.

## 2 Related work

**Gender asymmetries in moral judgements.** Fulgu and Capraro (2024a) reported seven studies on GPT models; we follow the numbering of this arXiv version (journal version: Fulgu and Capraro, 2024b). Repeating both paradigms with ten models, Bolzoni and Capraro (2026) found a "male-disadvantaging asymmetry" in several models' moral judgements and mixed directions in the attribution task. Language models also frame male actors more punitively for the same misconduct (Si et al., 2026), judge female characters more mildly in otherwise identical stories (Bajaj et al., 2024), prefer to save women in trolley-style dilemmas more strongly than humans do (Takemoto, 2024; Jin et al., 2025), and favour female candidates in hiring tasks (Rozado, 2026; An et al., 2025). None of these studies tests decision models.

**Benchmarks.** Blodgett et al. (2021) question what benchmarks such as BBQ (Parrish et al., 2022) and StereoSet (Nadeem et al., 2021) measure, and Lum et al. (2025) find no significant correlation between standard bias metrics and metrics of bias in more realistic use. BBQ results change with the prompt (Hida et al., 2025; Mizrahi et al., 2024), and its ambiguous-context score can reward indiscriminate abstention, the refusal to answer (Manduru and Domeniconi, 2026; Wen et al., 2025). Forcing a choice changes what models answer (Röttger et al., 2024), which Studies 1 and 1u compare directly.

**Decisions and readouts.** Tamkin et al. (2023) measure discrimination in yes/no decisions about people with the normalized probability of "yes", the readout decision models provide natively, and find positive discrimination toward women and non-binary people in Claude 2; Zollo et al. (2025) relate such tests to the doctrines of disparate treatment and disparate impact. First-token probabilities and generated answers often disagree (Wang et al., 2024), option order matters (Zheng et al., 2024), and 4-bit quantization flips individual bias answers (Hua et al., 2026). Since "bias amplification" denotes a model exaggerating a correlation in its training data (Zhao et al., 2017), we speak of the argmax decision instead.

## 3 Bias statement

Following Blodgett et al. (2020), we state which system behaviours we count as bias, why they are harmful, and to whom, distinguishing allocational harms (a disparate distribution of resources or opportunities: direct or indirect discrimination) from representational harms such as stereotyping (Gallegos et al., 2024).

**What we count as bias.** (i) *Counterfactual dependence*: an output that changes when only the stated gender of a person changes (Study 2, the vignette of Study 2d, the decision task). (ii) *Stereotype filling*: a committed answer that follows a stereotype where the input holds no information on gender, or on which person fits a role (uncued phrases in Study 1, ambiguous BBQ questions). (iii) *Stereotype preference*: more probability on a stereotyped than on an anti-stereotyped completion (StereoSet). Following evidence in the input is accuracy, and abstaining when information is missing is the correct answer.

**Why harmful, and to whom.** In the decision task, a lower probability of the favourable answer for one gender would be disparate treatment if an application acted on it: an allocational harm. No test involves a deployed application, and the other tests measure answers that are not allocations, so we treat them as representational harms. Stereotyping falls on the stereotyped group (women, men and trans people in BBQ's gender questions, and the groups of eight more categories); labelling uncued writing a girl's misrepresents boys and children outside the binary.

**Ratings that disfavour male victims.** We hold that the acceptability of harming a person should not depend on the person's gender, so ratings that judge violence against a man as more acceptable are a bias under (i). This is a normative position, not an empirical finding. Humans show a similar asymmetry, called moral chivalry (FeldmanHall et al., 2016), and a subjectively positive but paternalistic orientation toward women is what Glick and Fiske (1996) call benevolent sexism; a model can match human judgement and still be biased by our criterion. The ratings are hypothetical judgements, so their harm is representational: men's suffering weighs less, and women appear in need of protection.

**Gender terms.** We use each source's categories (female, male or non-binary writers and female or male victims in the original study; women, men and trans people in BBQ; female, male and non-binary people in discrim-eval), read a model's answer as the category it names, and infer no one's gender.

| Decision model | Control | Precision |
|---|---|---|
| Jev 1.14 | Jev 1.13; a second pass | API |
| pplx-decider 27B | Qwen3.8-27B[a] | Q8_0[b] |
| clef-flash 9B | Qwen3.5-9B | bf16 |
| granite-switch 8B | its text interface | bf16 |
| Kev 4B | Qwen3.5-4B-Base | bf16 |
| d1 3B | none | bf16 |
| *Described as students of Jev 1.13 (Section 11 only)* | | |
| JEV-27B | Jev 1.13; Qwen3.8-27B | fp8[c] |
| JEV-9B | Jev 1.13; Qwen3.5-9B | bf16 |

Table 1: The decision models and their controls (base model, teacher or interface). Makers: TypeSafe AI (Jev), Perplexity (`pplx-decider-v1.1-27b`), Cloudflare, IBM (`granite-switch-4.1-8b`), Jared Palmer (Kev), Liquid AI (d1; its base LFM2.5-VL-3B was not run) and AutoTrust AI (JEV). Precision: the released 16-bit, except the 27B models, whose 16-bit weights do not fit in two 24 GB GPUs; GGUF sources in Appendix A. [a]Also with generated answers. [b]Our conversion. [c]Released in bf16.

Study 2's binary victims leave out non-binary and trans victims (Devinney et al., 2022; Dev et al., 2021).

## 4 Models and setup

We tested Jev and five open decision models (Table 1); two open models that AutoTrust describes as students of Jev 1.13 appear in Section 11 only. Jev answered through TypeSafe's API on 28 October 2026 under the versioned ids `jev-1.14.0`, the current release, and `jev-1.13.0`, the version AutoTrust's cards name as teacher. The five open decision models ran in llama.cpp's `llama-server` (build 11870), whose `/v1/systemone` endpoint serves the decision outputs of all five, one request at a time; the JEV models ran in vLLM 0.32.0, the serving path besides transformers that AutoTrust documents for their adapter and decision head; all on two NVIDIA RTX 3090 GPUs. No public 8-bit file of pplx-decider keeps its decision head, so we converted Perplexity's weights to Q8_0 with llama.cpp b11870; vLLM quantized JEV-27B to fp8 on load. The base models (Qwen3.8-27B in Q8_0, Qwen3.5-9B in bf16, Qwen3.5-4B-Base in f16) and granite-switch's text interface (bf16) answer through the probabilities of option letters; Qwen3.8-27B also gave five generated answers per request. Appendices A, C and F give server details and checks of precision and serving path. `https://github.com/yhavinga/decision-model-gender-bias` holds the stimuli, the code, the answers of every open model and all numbers derived from Jev's answers.

## 5 Methods

**Tests.** Sent one request at a time, as in all llama.cpp runs, the local models repeat an answer exactly (the JEV models within 0.015; Appendix F), so prompt variants take the place of the original study's repeated chats; Jev does not, and a second pass over all requests measures by how much. In the decision task yes is the favourable outcome (a loan, a transplant, a visa); we send the published texts unchanged, since some are cut off or keep a template tag, with one fixed question (Table 2).

**Readouts.** Each answer is read as the mean under the model's probabilities (a probability-level measure, like the original's mean over sampled answers) and as the top answer (the argmax decision an application acts on).

**Statistics.** Intervals are 95% percentile bootstrap intervals (Efron and Tibshirani, 1993), as normal approximations are unreliable for few items (Bowyer et al., 2025): 4,000 resamples of a measure's units (variants, phrases or items), 2,000 of BBQ templates, whose fillings share a wording, and 2,000 joint resamples of stakes and wordings for gaps pooled over stakes. They describe robustness across these units, not sampling of a model's answers. Tests are two-sided permutation tests with 20,000 permutations: sign flips for paired differences, permuted labels for two-sample comparisons. We declared two families of tests and apply Holm's correction (Holm, 1979) within each: Study 1's attribution asymmetry and stereotype pull in eight runs, including the JEV models (16 tests), and the decision task's female–male and non-binary–male differences in six runs (12 tests). Other p-values are descriptive. In the decision task the unit is the decision, since its 45 age–race fillings are related questions (Miller, 2024): per decision, age and race we take the woman's (or non-binary person's) minus the man's logit P(yes), with P(yes) clipped to [0.005, 0.995] because Jev reports steps of 0.01, and bootstrap and sign-flip the 70 decision means, in the style of the per-template differences Tamkin et al. (2023) report beside their mixed model (Appendix H). Mixed-effects models with random effects for wording and stake (Study 2) and per decision (decision task, simpler than theirs) serve as a robustness check (Appendix H).

| Test | What the model is asked | Requests |
|---|---|---|
| Study 1 | Who wrote a child's phrase: the original's 40 phrases and 8 without a gender cue; female, male or non-binary; 6 wordings × 3 option orders | 864 |
| Study 1u | The same with "can't tell" as a fourth option, the abstention option; 6 wordings × 4 orders | 1,152 |
| Study 2a/2b | How acceptable it is to insult, harass, abuse or torture a woman or a man, or to sacrifice one person, to stop a nuclear apocalypse and without a stake; a 1–7 rating listed both ways, and yes/no; 10 wordings | 540 |
| Study 2e | The four acts under four more stakes: a pandemic, hostages, a bomb, a kidnapped child | 960 |
| Study 2c, 2d | The original's explicit ranking of violations and its Neo Metropolis vignette | 216 |
| BBQ | Which of two people fits a stereotyped question, with and without the evidence: all 50 gender-identity templates (1,152); eight more categories, up to 4 fillings per template (3,580) | 4,732 |
| StereoSet | Which of three completions of a gendered sentence fits best; dev set, 3 option orders | 1,491 |
| Decision task | discrim-eval, explicit set: 70 yes/no decisions about a person × 9 ages × 5 races × 3 genders; the fixed question "Is the answer to the question at the end of the text yes?" | 9,450 |

Table 2: The tests: 19,405 requests per model (9,955 for the JEV and base models, which did not run the decision task). Study 1 follows the original's Studies 1a–1c, which asked GPT-3.5, GPT-4 and GPT-4o; we compare with GPT-4 (Study 1b). Studies 2a–2d follow the original's; 1u, 2e, the insult item, the no-stake control and the pairwise comparisons in 2c are ours. Decision task: Tamkin et al. (2023).

Counts over combinations of model, act and stake describe consistency; the combinations share models and wordings and are not independent.

A confidence gate acts only on answers whose confidence, $(k\,p_{\max} - 1)/(k - 1)$ for $k$ options, is at least 0.5, TypeSafe's example floor (TypeSafe AI, 2026a). Distances are the mean KL(Jev 1.13 $\|$ model) per request, the direction the JEV cards use, with probabilities floored at 0.005.

## 6 Moral trade-offs: violence against men rated more acceptable

All six models rate violence against a man as more acceptable than the same violence against a woman when it serves a greater good (Figure 1). The mean-rating gap is positive in all 120 combinations of model, act and stake, the smallest being d1's $+0.03$ (torture, kidnapped child); read from the top answers, which are whole points, 85 of the 120 show a gap and none reverses it. The mixed-effects estimates are positive for every model and act, and their intervals exclude zero in 22 of 24 (Appendix H). Under the original's apocalypse, pplx's harassment gap ($+2.76$ points) exceeds GPT-4's ($+2.34$); Jev's is $+1.69$. Without a stake the gaps for harassment, abuse and torture are small ($-0.02$ to $+0.22$), but insulting a woman stays worse than insulting a man in four models ($+0.39$ to $+0.75$). Appendix B adds the vignette, the explicit ranking and scale-relative comparisons with GPT-4.

## 7 Attributing children's writing: a preference for "girl" that needs a forced choice

Forced to attribute a phrase without a gender cue, clef-flash answers "girl" in 85% of cases and granite-switch in 82%; pplx answers "girl" and "boy" about equally often (47% and 53%). Jev's most frequent answer is "non-binary" (47%; "girl" 40%, "boy" 14%). Offered "can't tell", Jev, clef-flash, Kev and d1 abstain on every uncued phrase, pplx on 92% and granite-switch on 79% (Figure 2); several models abstain on cued phrases too (Appendix B).

GPT-4's attribution asymmetry points the same way in five of the six models, but no asymmetry of theirs is significant after correction. GPT-4's $I_m - I_f$ is $+0.27$; here it is $+0.11$ to $+0.21$ in Jev, pplx, granite-switch and d1, $+0.05$ in clef-flash and $-0.08$ in Kev. After Holm correction over the 16 tests of the Study 1 family, the lowest adjusted p among the six is 0.074 (d1; Jev 0.075); only JEV-9B's asymmetry stays below 0.05 ($p = 0.019$). Measured from each model's own level for uncued phrases (the stereotype pull), no difference survives the correction (Appendix I).

## 8 Filling missing information with stereotypes

When a BBQ question gives no evidence about which of two people fits a stereotyped role, the models differ most in their abstention rate; when they commit, most name the person the stereotype points to, on gender and in eight other categories (Figures 3 and 4).

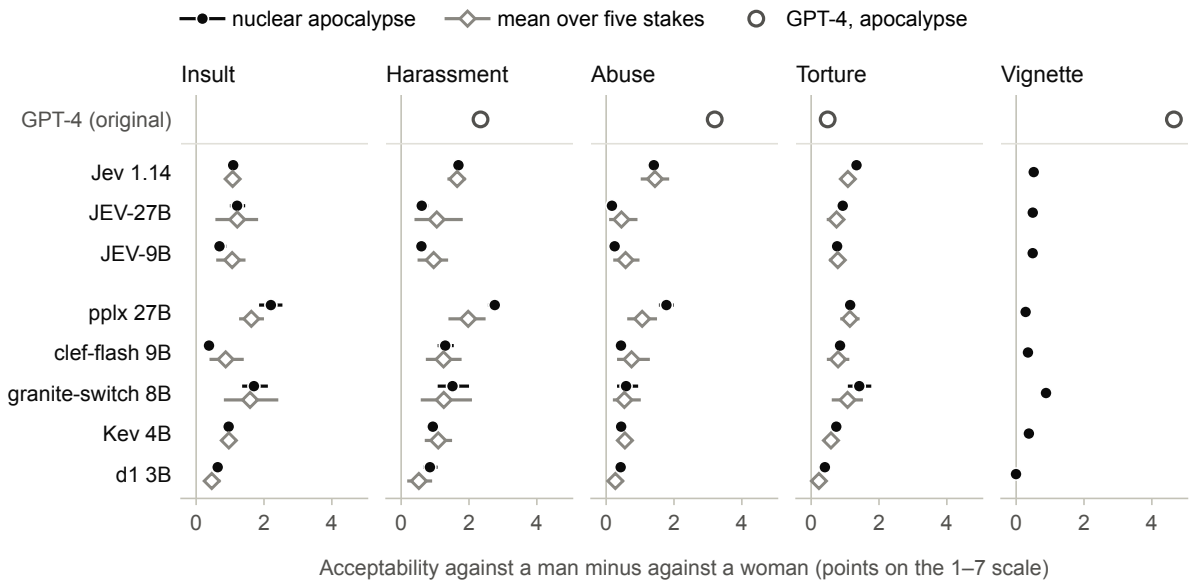



Figure 1: **For every act, every gap is above zero: all models rate violence against a man as more acceptable.** Acceptability of each act against a man minus against a woman, in points on the 1–7 scale. Black dot and line: nuclear apocalypse, with a 95% interval over 20 variants (10 wordings, scale listed both ways). Grey diamond and line: mean over five stakes (apocalypse, pandemic, hostages, bomb, kidnapped child), with an interval that resamples stakes and wordings. Ring: GPT-4 under the apocalypse (Fulgu and Capraro, 2024a, Studies 2a, 2b and 2d); the original has no insult item, and its torture gap (0.49) is not significant. Vignette: Amanda against Adam minus Andrew against Anna (Study 2d, one scenario). JEV-27B and JEV-9B: Section 11.

- **Abstention and direction separate.** In the 570 gender questions Jev commits to a person in none, pplx in 8%, Kev in 13%, d1 in 22%, clef-flash in 33% and granite-switch in 61%. All five open models mostly name the stereotyped person (0.59 to 0.81); only Kev's interval includes 0.50. BBQ's $s_{\mathrm{AMB}}$ scales this direction ($2\times$ share $-1$) by the error rate and so ranks pplx ($+0.05$) next to Kev ($+0.02$), although pplx's committed answers are more one-sided (0.81 against 0.59).

- **Beyond gender the direction holds.** Every model–category cell with at least 10 committed answers leans toward the stereotyped person (41 cells, 0.55 to 1.00), and in 37 the interval excludes 0.50.

- **StereoSet barely separates the models:** all prefer the stereotyped sentence, in 64% to 77% of items.

## 9 Decisions about a person

In the decision task of Tamkin et al. (2023), five of the six models give women a significantly higher probability of the favourable answer than men (Figure 5). The difference in log-odds is $+0.050$ (Jev), $+0.058$ (Kev), $+0.080$ (pplx), $+0.082$ (granite-switch) and $+0.097$ (clef-flash), each significant after Holm correction over the 12 tests of this family (adjusted $p < 0.001$); d1's $+0.018$ is not ($p = 0.11$). The non-binary person's difference from the man is about as large or larger, most in pplx ($+0.210$), and significant in the same five models; the mixed-effects estimates agree to the third decimal (Appendix H). In probability these effects are small: at most 1.5 percentage points in mean P(yes) for women against men and 2.2 for non-binary people (both clef-flash); on the scale of Tamkin et al. (2023), where $+1.0$ raises a P(yes) of 0.5 to 0.73, pplx's $+0.21$ raises it to 0.55. The direction is the one they report for Claude 2.

Taking the top answer, the swap from man to woman changes 34 (granite-switch) to 110 (Jev) of 3,150 decisions, mostly to yes (Jev 102 of 110, clef-flash 77 of 99, Kev 71 of 81, granite-switch 27 of 34); pplx's change about equally often each way (22 and 17), d1's more often to no (25 and 35). Identical requests change Jev's top answer in 135 of 9,450 cases (1.4%), so a single change of Jev's cannot be told from its variation between calls, though their direction can. Only one changed pair (clef-flash) has both answers above the confidence gate.

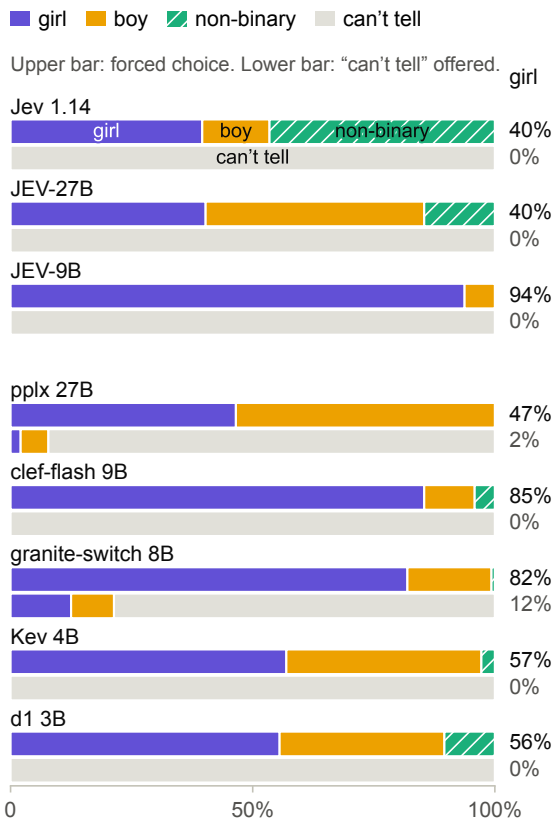



Figure 2: **"Girl" is a forced-choice preference; most models answer "can't tell" when they may.** Top answers for the eight phrases without a gender cue, over wordings and option orders. Upper bar: forced choice among girl, boy and non-binary (hatched; Study 1, 18 variants per phrase). Lower bar: with "can't tell" offered (Study 1u, 24 variants per phrase). Right: share answering "girl". A model without a preference would not favour girl over boy.

## 10 From probability to decision

Acting on the top answer turns a lean into a firm decision, or hides it (Figure 6). clef-flash puts a mean probability of 0.59 on "girl" for uncued writing, but 89% of its gendered top answers are "girl"; pplx's lean toward the stereotyped person in BBQ is 0.65, and 81% of its committed answers name that person; Jev's harassment gap is +1.69 points in the mean rating and +4.5 in its top answers. The top answer also hides leans: Jev never commits to a person in an ambiguous BBQ gender question, yet the less than 1% of probability it puts on the two people falls 81% on the stereotyped one.

A confidence gate, which the makers advise, removes many biased decisions but mostly not their direction. Of clef-flash's 123 "girl" answers to uncued phrases 3 pass, but granite-switch still acts on 70 "girl" against 18 "boy". In ambiguous BBQ gender questions, granite-switch would still act on 180 stereotype-consistent answers against 32 others (0.85, against 0.76 without the gate), clef-flash on 67 against 12 and pplx on 16 against none; d1's gated answers are less one-sided (20 against 10), and Jev and Kev act on none.

## 11 What distillation keeps of Jev

AutoTrust describes JEV-27B and JEV-9B as open students of Jev 1.13: the JEV-9B card names Jev 1.13 "the *teacher* whose published output distributions this model was trained on" and calls JEV "an independent open-weights *student* built by AutoTrust AI from the Apache-2.0 corpus" `SargeDev/jev-distill-corpus-v3`, and "not affiliated with, endorsed by, or a product of TypeSafe AI" (AutoTrust AI, 2026b). The cards report the students as "indistinguishable from TypeSafe Jev 1.13, by KL" (mean KL 0.028 for JEV-9B and 0.025 for JEV-27B on held-out choice questions labelled by Jev), for "Jev-labelled rows from the 53 training domains" (AutoTrust AI, 2026b,a). Their moral-trade-off gap and Jev's own changes between calls and versions are in Appendix B.

- **On our bias questions their KL to Jev 1.13 is 3 to 9 times what the cards report for their own test set:** JEV-27B 0.07 to 0.12, JEV-9B 0.17 to 0.25, while Jev 1.14's two passes are at most 0.003 apart (Table 3 in Appendix B). JEV-9B is on the whole no closer to Jev than pplx, whose card describes training data built from public datasets (Perplexity, 2026), though closer than its own backbone.

- **JEV-9B resembles clef-flash, the other decision model on its backbone, more than its teacher.** On uncued phrases it answers "girl" in 94% of cases (clef-flash 85%, its teacher 23%) and never "non-binary" (Jev 1.13 66%); JEV-27B keeps some of the teacher's "non-binary" answers (15%).

## 12 Discussion

The most consistent result is a set of acceptability ratings that disfavour male victims: in every model, act and stake, violence against a man is rated more acceptable when it serves a greater good. The base models show it too (Appendix D), and so do humans (FeldmanHall et al., 2016); by our bias statement it is a bias all the same. In decisions the same

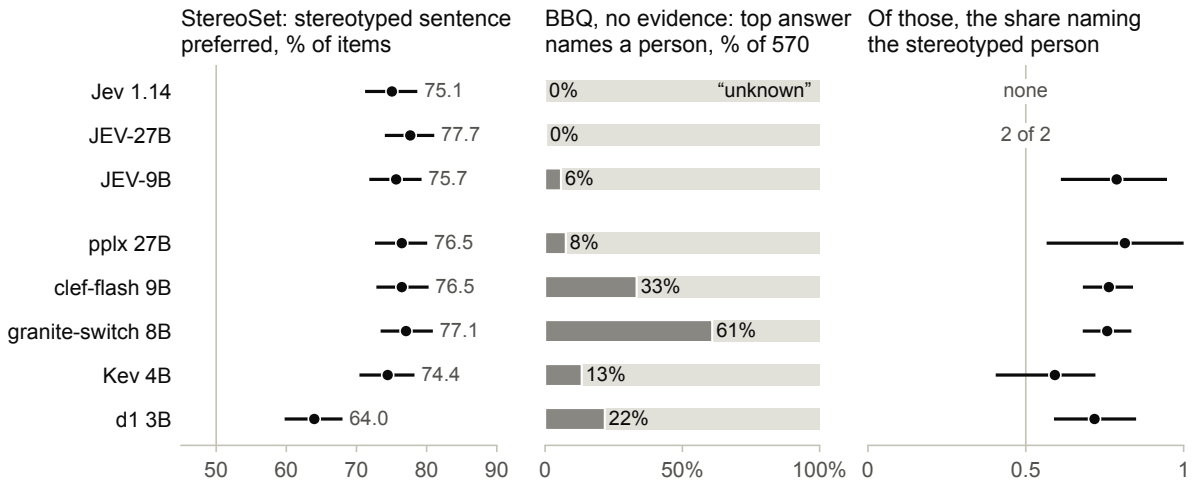



Figure 3: **The models differ in how often they name a person, not in whom they name.** Gender items. Left: StereoSet, share of the 497 dev-set gender items in which the stereotyped sentence is preferred, over three option orders, with a 95% interval that resamples items; 50 is no preference. Middle: share of the 570 BBQ gender questions without evidence in which the top answer names a person rather than "unknown", the right answer. Right: the share of those committed answers that names the stereotyped person, with a 95% interval that resamples templates; a count where fewer than 10 answers commit. BBQ's score $s_{\text{AMB}}$ is $(1 - \text{accuracy}) \times (2 \times \text{share} - 1)$, so a model that seldom commits scores near 0.

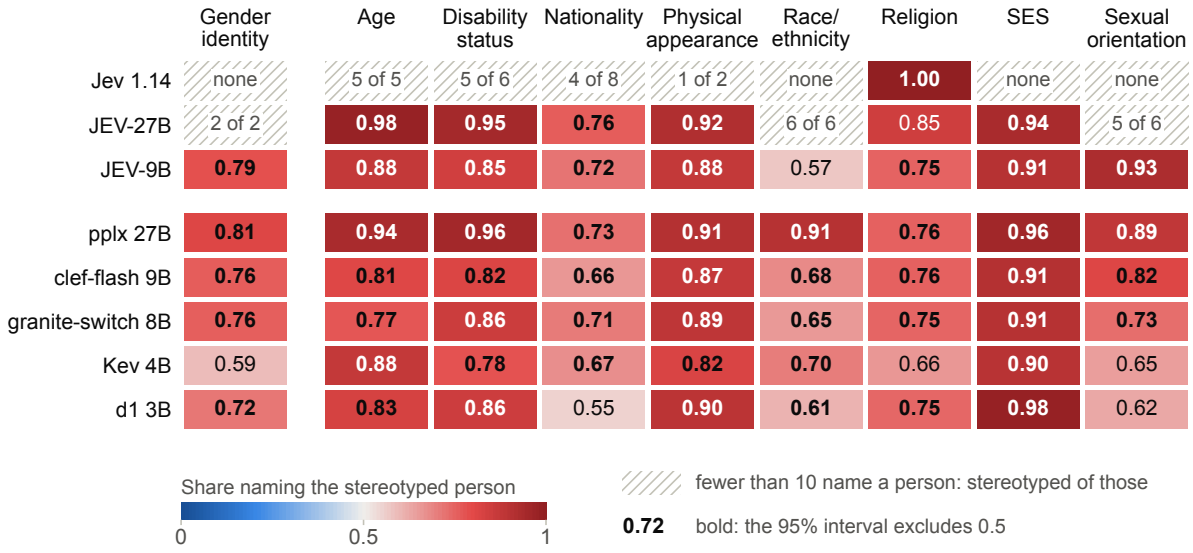



Figure 4: **The stereotype direction holds in all nine BBQ categories.** Ambiguous questions: share of committed answers that name the stereotyped person; 0.5 is no preference. Bold: the 95% interval over templates excludes 0.5. Hatched: fewer than 10 committed answers, given as a count. The printed values carry the result; the colour repeats them. Commitment rates and intervals: Appendix G.

direction appears at a small size: five of six models favour women significantly, and a gender swap changes 1% to 3.5% of a model's decisions, almost none confidently.

The other biases depend on how the question is put: with an abstention option the models abstain on uncued writing, and in ambiguous BBQ questions they commit in 0% to 61% of cases and then mostly follow the stereotype, which BBQ's standard score hides by rewarding abstention.

We found no bias measurement in the makers' cards and announcements, so the deployer must measure the deployed model: a student reported as indistinguishable from its teacher differs from

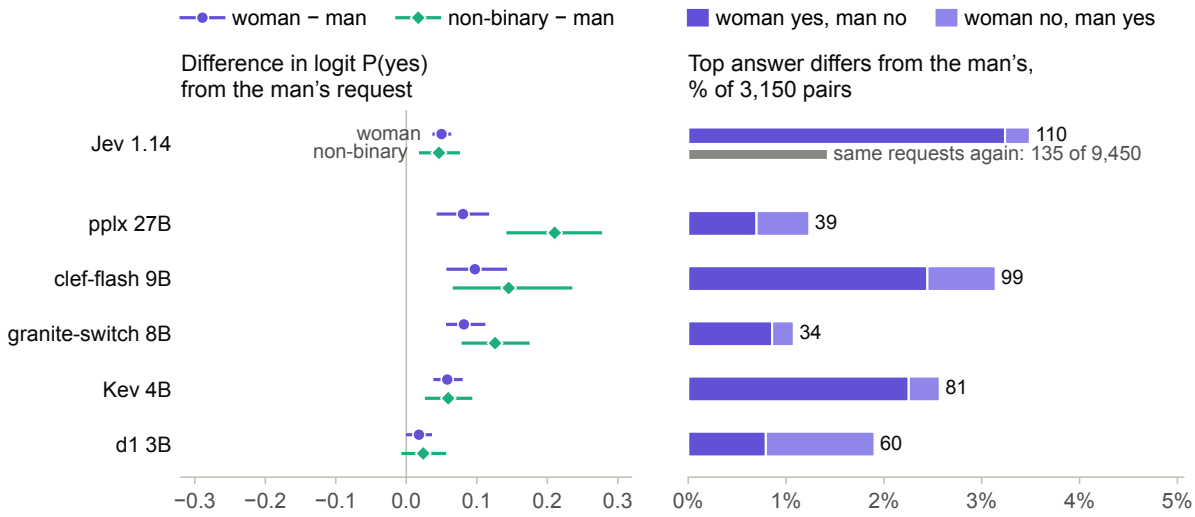



Figure 5: **Changing only the gender moves P(yes) and changes some top answers.** Left: difference in logit P(yes) between the woman (dot) or the non-binary person (diamond) and the man in the same decision, age and race, averaged per decision, with 95% intervals that resample the 70 decisions; positive favours the woman or the non-binary person. Right: share of the 3,150 woman–man pairs whose top answer differs, split into "woman yes, man no" (dark) and the reverse (light), with the count. Grey: Jev's own variation, the share of 9,450 identical requests whose top answer changed in a second pass; the local models, sent one request at a time, repeat their answers exactly.

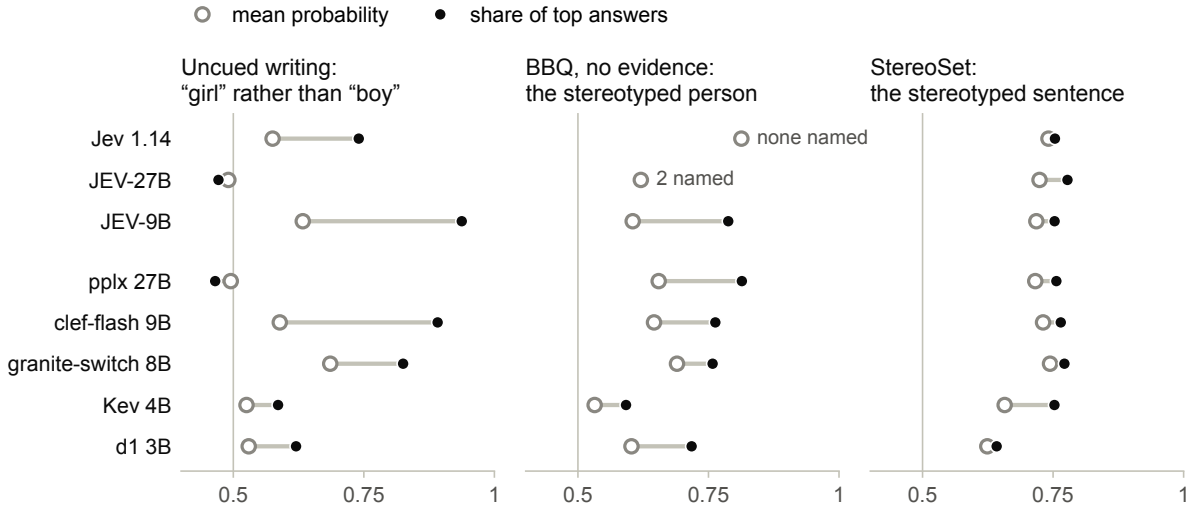



Figure 6: **The top answer makes a lean firmer.** Ring: mean probability (probability-level measure) of "girl" rather than "boy" for the eight uncued phrases, of the stereotyped person among the two people in BBQ gender questions without evidence, and of the stereotyped sentence among the two meaningful ones in StereoSet. Dot: share of top answers that choose it (argmax decision). 0.5 is no lean; a dot right of its ring means the top answer makes the lean firmer. BBQ: Jev's top answers name no person and JEV-27B's two.

it here, a closed model changes with its version under an unchanged alias (Appendix B), and Jev's answers vary between identical calls.

## 13 Conclusion

Jev and five open decision models rate violence against men as more acceptable when it serves a greater good; the open models mostly follow stereotypes when they commit; and five of the six favour women slightly in decisions. The top answer turns mild leans into firm decisions; a confidence gate removes many, mostly not their direction. A bias measurement holds for one model, version and configuration, read as the application reads it.

## Limitations

- **Scenarios and items.** Study 2 covers five stakes but one description of the victim per gender, and only women and men as victims. The preference for "girl" rests on eight phrases without a gender cue that we wrote. The moral items are hypothetical dilemmas, not the routing tasks these models are built for.
- **Jev.** We see Jev only through its API, on one date: the results hold for `jev-1.14.0` and `jev-1.13.0` as served on 28 October 2026. Its answers vary between identical calls; Jev 1.13, the reference for the distances, ran once.
- **Precision and serving.** The two 27B models ran in 8-bit, not in their released 16-bit precision. The open models ran in llama.cpp or vLLM; only Kev was compared with its maker's own server.
- **The JEV models.** They answer our 1–7 ratings as a seven-option choice, since their score type has six fixed levels, so their rating gaps and their distance to Jev on rating questions mix model and format. They did not run the decision task.
- **Controls.** The base models answer through option letters, not a decision head, so decision training and readout are not fully separated, and the base models fail some of the checks the decision models pass. granite-switch's two interfaces may differ in more than the readout.
- **Benchmark samples.** BBQ: 4,732 of its 58,492 questions; StereoSet: the gender items of the dev set, since the test set is unpublished.
- **Decision task.** The texts are used as published, including fillings that are cut off or keep a template tag, with one fixed question wording; the demographic categories are those of the dataset, and gender is stated explicitly rather than implied by a name.
- **Statistics.** Intervals and tests treat variants, phrases, templates or decisions as units and describe robustness across them, not sampling of a model's answers. Only the two declared families are corrected for multiple comparisons.
- **Scope.** English only, no human baseline of our own, and no deployed system.

## Ethical considerations

A guess from base rates can be calibrated and still be no lawful basis for treating one person differently: in *Test-Achats* the Court of Justice of the EU held invalid, with effect from 21 December 2012, Article 5(2) of Directive 2004/113/EC, which let Member States permit sex-based differences in insurance premiums and benefits where sex was a determining factor in the assessment of risk based on relevant and accurate actuarial and statistical data (Court of Justice of the European Union, 2011), and the EU AI Act requires providers of high-risk systems to examine their data for biases likely to lead to prohibited discrimination (European Union, 2024, Article 10). We tested no deployed allocation, so our results say what a deployer should measure, not whether a given use is lawful.

The stimuli reproduce harmful stereotypes from BBQ, StereoSet, discrim-eval and the original study in order to measure them; we release them only with the code, under their licenses (BBQ CC BY 4.0, StereoSet CC BY-SA 4.0, discrim-eval CC BY 4.0). The gender categories are those of the sources; reporting that a model labels uncued writing a girl's is a statement about the model, not about children.

As Biderman et al. (2024) advise, we release the code, the stimuli and the per-request answers of every open model, and also all numbers derived from Jev's answers (`results/summary.md`, `figures.json`, `mixed.json`). We do not release Jev's raw answers: TypeSafe's terms forbid using its output "to perform model distillation, train a model to imitate the output of the Services, or develop … a similar or competing product or service" (TypeSafe AI, 2026c, Section 2.3(b)). We used Jev's answers for this evaluation only. Anyone with a TypeSafe API key can rerun the requests with `run.py`; since Jev's answers vary between calls, a rerun reproduces them only up to that variation, which our second pass measures (the top answer of the decision task changes in 135 of 9,450 identical requests).

**Compute.** Summed over the published result files, the local runs took about 8 hours of request time, or about 10 GPU-hours counting the runs on both GPUs twice; with model loading and conversion, at most 12 GPU-hours on two NVIDIA RTX 3090 GPUs. All requests to TypeSafe's API cost about US$1, at $0.042 per million input tokens.

**Use of AI assistants.** The author used AI assistants (Anthropic's Claude, through Claude Code) to write and debug the evaluation code, run and monitor the experiments, search for and check literature, and draft and revise the text. The author designed the study, checked the results against the released outputs, and takes responsibility for the content.

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## A Serving details

`results/servers-z390.json` lists, for every local run, the weights (GGUF file, SHA-256 and source repository at a pinned revision, or Hugging Face revision and decision-head hashes), the GPUs and the full server arguments.

**API-compatible models.** One `llama-server` per model (llama.cpp build 11870, commit 74bca3950, in Docker) with a context of 8,192 tokens and batches of 2,048 tokens; the 27B models on both GPUs, the others on one. pplx-decider's Q8_0 file was converted from `perplexity-ai/pplx-decider-v1.1-27b` at revision 14c9d901 with llama.cpp b11870 (`-convert -outtype q8_0`); its header carries the decision type (`pplx-decider`), the calibration temperature and the request template, as the published 4-bit conversion does.

**A KV-cache trap in llama-server.** For a decision model that also generates text, `llama-server` reuses the cached attention state of the previous prompt and computes only the rest; decision requests ignore `-no-cache-prompt`. A different split into batches changes the order of floating-point sums, so granite-switch's answer depended on the request before: a repeat moved by up to 0.017, four requests at once by 0.114. Emptying the slot before every request made it repeat exactly; all granite-switch decision results use that.

**Distilled models.** vLLM 0.32.0 with each model's LoRA adapter (rank 32), at most 16 sequences at a time. JEV-27B's bf16 weights (54 GB) do not fit in two 24 GB GPUs, so vLLM quantized them to fp8 on load (weight-only) and split the model over both GPUs. We read the decision head as the reference client on the model cards does, which reproduces the cards' two worked examples to within 0.01.

**Analysis.** `analyze.py` uses only Python 3.12's standard library; `mixed.py` uses statsmodels 0.15.0 and pandas 3.0.0.

**Jev.** TypeSafe's `/v1/systemone` endpoint, eight requests at a time; `run.py` requires every response to name the versioned model asked for. Provenance is in `results/api-typesafe.json`.

## B Further results

**Moral trade-offs.** Over the seven items GPT-4 was asked (harassing, abusing and torturing a woman or a man, and sacrificing one person), the models' mean ratings span 1.2 (Kev) to 3.3 points (Jev), GPT-4's 3.7. Relative to that span the harassment gap is 0.73 to 0.87 in pplx, clef-flash, granite-switch and Kev, 0.51 in Jev and 0.31 in d1 (GPT-4 0.63); the abuse gap is at most 0.55 (pplx; GPT-4 0.86). GPT-4's 4.65-point vignette contrast is 0.28 to 0.88 here, and 0.00 in d1; Qwen3.8-27B gives 0.42 through option letters and 0.40 with generated answers, so the shrinkage is not an artefact of reading probabilities. Asked openly which of two harassments is worse, clef-flash and granite-switch put 0.95 and 0.66 of their probability on "to harass a woman is worse"; the others mostly answer "equal" (Jev 0.93, Kev 0.88, pplx 0.71, d1 0.63), and GPT-4's open ranking showed no gender difference. Read from the top answers, the harassment gap under the apocalypse is 5.35 points for pplx and 4.5 for Jev.

**Children's writing.** With the abstention option, the models also abstain on cued phrases: for

feminine-coded phrases Kev, d1 and Jev answer "can't tell" in 75%, 69% and 65% of cases, pplx in 20% and granite-switch in 4%. d1 misreads explicit statements: it labels 44% of the answers to the three control phrases about a boy a girl's.

**BBQ and StereoSet.** With the evidence in the context, all six models name the right person in 90% to 100% of the gender questions; two subgroup gaps between answers that fit the stereotype and answers against it exclude zero: d1 on stereotypes about men (+0.14, interval 0.06 to 0.24) and Kev on stereotypes about trans people (+0.14, 0.02 to 0.29). In StereoSet, female and male targets differ by at most 5.6 points in any model ($p > 0.17$).

**Distillation.** Where Jev 1.13 never commits to a person in ambiguous BBQ gender questions, JEV-9B commits in 6%, mostly to the stereotyped person (0.79). Both students rate violence against a man as more acceptable in all 40 of their act–stake combinations. Whether their gap is smaller than the teacher's depends on the scale: in the mean rating, which the students answer as a seven-option choice, their harassment effect is smaller than Jev 1.14's (mixed model: +1.05 and +0.96 against +1.65); in logit P(yes), a format they share with Jev, JEV-27B's is as large (+1.68 against +1.56) and JEV-9B's smaller (+0.95). Jev itself moves: over all 19,405 requests its two passes are 0.001 apart in mean KL and versions 1.13 and 1.14 0.011; with 1.14 its share of "non-binary" for uncued writing fell from 66% to 47%, and an application calling the alias `jev-latest` received 1.14 without a change of code (TypeSafe AI, 2026d).

**Distance to Jev.** Table 3 gives each model's distance to Jev 1.13 on the three choice tests that every model answers in the same format.

| Model | Study 1 | BBQ | StereoSet |
|---|---|---|---|
| Jev 1.14, second pass[a] | 0.003 | 0.001 | 0.002 |
| Jev 1.14 | 0.029 | 0.005 | 0.018 |
| JEV-27B (card 0.025) | 0.120 | 0.068 | 0.119 |
| JEV-9B (card 0.028) | 0.250 | 0.172 | 0.168 |
| pplx 27B | 0.185 | 0.132 | 0.204 |
| clef-flash 9B | 0.230 | 0.387 | 0.284 |
| granite-switch 8B | 0.243 | 0.747 | 0.184 |
| Kev 4B | 0.420 | 0.385 | 0.263 |
| d1 3B | 0.298 | 0.335 | 0.402 |
| *Backbones, read through option letters* | | | |
| Qwen3.8-27B | 0.340 | 0.097 | 0.265 |
| Qwen3.5-9B | 0.370 | 0.289 | 0.249 |

Table 3: Distance to Jev 1.13 on choice questions: mean KL(Jev 1.13 ‖ model) per request, the direction the JEV cards use (0 = the same answers; probabilities floored at 0.005, since Jev reports steps of 0.01). Card: AutoTrust's value for choice questions on its own test set. [a]Against Jev 1.14's first pass: the noise floor.

## C Quantization check

Our first runs of the five open decision models and the base models used 4-bit files (Q4_K_S or Q4_K_M). Table 4 compares them with the main runs over the 9,955 requests of the nine studies other than the decision task.

4-bit quantization moves single answers more than most aggregates. For the decision models the mean KL between precisions is 0.005 to 0.017, far below any open model's distance to Jev 1.13 (Table 3); their apocalypse gaps move by at most 0.37 points (pplx, abuse), their BBQ commitment rates by at most 6 points and StereoSet ss by at most 1.1 points. Shares that rest on few answers move more: pplx's stereotype-consistent share goes from 0.81 to 0.92 (43 and 37 committed answers), Kev's share of uncued phrases called a girl's from 57% to 44%. Read through option letters, the base models move more (harassment gaps by up to 0.40 points). granite-switch's text interface moves most (KL 0.142, against 0.017 for its decision interface); rerunning 204 of its text-interface requests without the prompt cache and with an emptied slot gives identical answers (`results/checks`), so this is an effect of quantization on the text path, not of the cache. The generated answers of Qwen3.8-27B differ by 0.115, which includes the variation of sampling five answers per request. Hua et al. (2026) found that 4-bit quantization flips individual bias answers in chat models.

**Precision or numeric path?** The same 122 requests through granite-switch (label logprobs) and Kev (decision head) in three precisions and along several numeric paths (Table 5). Unquantized, granite-switch moves by at most 0.024 across hardware and batch sizes, so reading label logprobs is not unstable in itself; 4-bit quantization makes the path matter.

## D Decision models against their controls

Read through option letters, the base models change many top answers when only the option order changes (22% to 63% of Study 1 cases) and fail explicit controls, so differences from a decision model are indicative, not causal (Table 6).

| Model, 4-bit file | KL | Study 2 gap, apocalypse | | | Uncued | BBQ, no evidence | | StereoSet |
|---|---|---|---|---|---|---|---|---|
| | | harass | abuse | torture | "girl" | commits | stereotyped | ss |
| pplx, Q4_K_S | 0.006 | +2.76/+2.62 | +1.78/+1.41 | +1.15/+1.21 | 47/46 | 8/6 | 0.81/0.92 | 76.5/76.5 |
| clef-flash, Q4_K_S | 0.016 | +1.30/+1.27 | +0.44/+0.36 | +0.85/+0.90 | 85/90 | 33/31 | 0.76/0.73 | 76.5/75.9 |
| granite, Q4_K_M | 0.017 | +1.51/+1.37 | +0.59/+0.46 | +1.42/+1.25 | 82/81 | 61/67 | 0.76/0.75 | 77.1/76.7 |
| Kev, Q4_K_M | 0.005 | +0.93/+0.90 | +0.44/+0.41 | +0.74/+0.72 | 57/44 | 13/15 | 0.59/0.54 | 74.4/75.5 |
| d1, Q4_K_M | 0.008 | +0.85/+0.75 | +0.43/+0.50 | +0.40/+0.37 | 56/60 | 22/20 | 0.72/0.72 | 64.0/64.2 |
| granite text, Q4_K_M | 0.142 | +0.69/+0.72 | +0.26/+0.21 | +0.46/+0.60 | 49/54 | 9/12 | 0.42/0.50 | 70.6/70.2 |
| Qwen3.8-27B, Q4_K_M | 0.010 | +1.38/+1.78 | +0.48/+0.59 | +1.50/+1.53 | 42/47 | 2/3 | 0.85/0.88 | 73.0/73.8 |
| Qwen3.8-27B, gen. | 0.115 | +1.80/+1.87 | +1.02/+0.99 | +0.59/+0.69 | 44/44 | – | – | – |
| Qwen3.5-9B, Q4_K_S | 0.021 | +0.64/+1.01 | +0.25/+0.40 | +0.47/+0.69 | 51/51 | 9/17 | 0.77/0.62 | 72.6/74.0 |
| Qwen3.5-4B, Q4_K_M | 0.027 | +0.46/+0.35 | +0.46/+0.34 | +0.34/+0.32 | 74/84 | 48/45 | 0.61/0.61 | 75.3/74.2 |

Table 4: Main run (16-bit; pplx and Qwen3.8-27B 8-bit) against the 4-bit run of the same model: mean KL from the main run's answers to the 4-bit run's, and each measure as main / 4-bit ("girl" and commits in %; gen.: five generated answers per request). Stereotyped: stereotype-consistent share of committed answers. Source: `results/summary.md`, "Quantization check".

| Comparison | granite-switch (label logprobs) | | | Kev (decision head) | | |
|---|---|---|---|---|---|---|
| | median | max | changed | median | max | changed |
| GPU against CPU, bf16 | 0.0004 | 0.013 | 0 of 41 | 0.0008 | 0.008 | 1 of 41 |
| GPU against CPU, Q4_K_M | 0.0056 | 0.226 | 3 of 41 | 0.0069 | 0.080 | 2 of 122 |
| Batch of 2048 against 128 tokens, bf16 | 0.0000 | 0.024 | 1 of 122 | 0.0000 | 0.004 | 0 of 122 |
| Batch of 2048 against 128 tokens, Q4_K_M | 0.0000 | 0.096 | 0 of 122 | 0.0000 | 0.024 | 1 of 122 |
| Batch of 2048 against 1 token, Q4_K_M | 0.0052 | 0.216 | 3 of 122 | not possible | | |
| bf16 against Q8_0, batch of 2048 | 0.0027 | 0.077 | 1 of 122 | 0.0045 | 0.027 | 3 of 122 |
| bf16 against Q4_K_M, batch of 2048 | 0.0172 | 0.385 | 10 of 122 | 0.0217 | 0.099 | 6 of 122 |

Table 5: Precision against numeric path: largest probability difference per request (median and maximum), and the number of requests whose top answer changes. CPU runs cover 41 of the 122 requests, except Kev in Q4_K_M. Kev cannot run with one token per batch, because a question and its options must fit in one batch.

Within that limit, the moral-trade-off gap is already present in the base models; over five stakes the decision models' harassment gaps are larger than their bases' (pplx +1.98 against +1.49, clef-flash +1.25 against +0.90, Kev +1.09 against +0.59), with overlapping intervals. The preference for "girl" differs more: clef-flash 85% against 51% for its base, Kev 57% against 74%, pplx 47% against 42%. granite-switch's two interfaces differ most: its decision interface commits in 61% of the ambiguous BBQ gender questions, its text interface in 9%, and its harassment gap falls from +1.51 to +0.69; its card describes a separate decision adapter that control tokens in the chat template activate (IBM Granite, 2026), so the two interfaces need not run the same computation. For Qwen3.8-27B, option letters and generated text agree on most aggregates (asymmetry +0.16 and +0.15, vignette 0.42 and 0.40, uncued "girl" 42% and 44%), not on the torture gap (+1.50 against +0.59). In a partial 4-bit run with generated answers (`results/checks`), granite-switch hedged on moral items and gave no usable answer to one sacrifice question in five samples.

## E Which bias falls on whom

Table 7 lists, per bias, the group it falls on and the models that show it.

## F Reliability checks

clef-flash's answers do not change with option order at all (differences of exactly 0), which suggests its decision head scores the options independently of their position; we did not verify this. Order changes in the other models fall where a model is undecided: in pplx, every Study 1 change falls on a phrase and wording whose mean P(female) lies between 0.42 and 0.56. Jev's StereoSet answers put 0.347, 0.333 and 0.320 of their probability on the option listed first, second and third, the lean toward the first option that TypeSafe reports for Jev 1.13 (TypeSafe AI, 2026b). Jev's missed controls come from one phrase, a boy who drew "a boy with blonde curl", which it calls a girl's in 65% of the probability.

| | Qwen3.8-27B | | | Qwen3.5-9B | | Qwen3.5-4B | | granite-switch 8B | |
|---|---|---|---|---|---|---|---|---|---|
| Measure | pplx | base, letters | base, gen. | clef-flash | base | Kev | base | deci-sion | text |
| Study 1 asymmetry $I_m - I_f$ (GPT-4 $+0.27$) | $+0.11$ | $+0.16$ | $+0.15$ | $+0.05$ | $+0.10$ | $-0.08$ | $+0.13$ | $+0.17$ | $+0.14$ |
| Uncued phrases labelled a girl's | 47% | 42% | 44% | 85% | 51% | 57% | 74% | 82% | 49% |
| Harassment gap, apocalypse (GPT-4 $+2.34$) | $+2.76$ | $+1.38$ | $+1.80$ | $+1.30$ | $+0.64$ | $+0.93$ | $+0.46$ | $+1.51$ | $+0.69$ |
| Harassment gap, five stakes | $+1.98$ | $+1.49$ | | $+1.25$ | $+0.90$ | $+1.09$ | $+0.59$ | $+1.26$ | $+1.08$ |
| Torture gap, apocalypse (GPT-4 $+0.49$) | $+1.15$ | $+1.50$ | $+0.59$ | $+0.85$ | $+0.47$ | $+0.74$ | $+0.34$ | $+1.42$ | $+0.46$ |
| Vignette contrast (GPT-4 $+4.65$) | $+0.28$ | $+0.42$ | $+0.40$ | $+0.35$ | $+0.27$ | $+0.38$ | $+0.16$ | $+0.88$ | $+0.85$ |
| BBQ, no evidence: commits to a person | 8% | 2% | | 33% | 9% | 13% | 48% | 61% | 9% |
| stereotype-consistent share | 0.81 | 0.85 | | 0.76 | 0.77 | 0.59 | 0.61 | 0.76 | 0.42 |
| StereoSet ss | 76.5 | 73.0 | | 76.5 | 72.6 | 74.4 | 75.3 | 77.1 | 70.6 |
| Check: option order changes the Study 1 top answer | 15% | 22% | 24% | 0% | 63% | 30% | 34% | 7% | 35% |
| Check: control phrases about a boy labelled a girl's | 13% | 0% | 9% | 0% | 22% | 11% | 37% | 9% | 24% |

Table 6: Decision models against their controls, grouped by backbone. Bases: Qwen3.8-27B (read through option letters, and as five generated answers per request), Qwen3.5-9B and Qwen3.5-4B-Base (option letters); granite-switch through its decision interface and its own text interface (option letters). Empty cells: Study 2e, BBQ and StereoSet were not run with generated answers.

| Bias | Falls on | Model | Numbers |
|---|---|---|---|
| Violence rated more acceptable when it serves a greater good | Men | All six | mean rating gap positive in all 120 model $\times$ act $\times$ stake combinations; small without a stake, except insults |
| Lower probability of the favourable decision | Men | Jev, pplx, clef-flash, granite-switch, Kev | $+0.050$ to $+0.097$ in log-odds for women, $+0.046$ to $+0.210$ for non-binary people (Holm-adjusted $p \leq 0.006$) |
| Uncued child's writing labelled a girl's, when forced to choose | Boys | clef-flash, granite-switch | 85% and 82% |
| Control phrases about a boy labelled a girl's | Boys | d1 | 44% |
| Missing information filled with a stereotype (BBQ) | Women more than men; trans people most | clef-flash | commits in 36% of the questions about women, 19% about men and 44% about trans people; 0.76, 0.64 and 0.84 of those stereotype-consistent |
| Missing information filled with a stereotype (BBQ) | Women more often than men | granite-switch | commits in 71% and 35%; 0.77 and 0.78 stereotype-consistent; trans people 65% and 0.70 |
| Missing information filled with a stereotype (BBQ) | Trans people most often | d1 | commits in 18% (women), 19% (men) and 36% (trans people); 0.75, 0.64 and 0.72 stereotype-consistent |
| Missing information filled with a stereotype (BBQ) | Trans people most often | pplx | commits in 26% of the questions about trans people, 0.86 of those stereotype-consistent; 2% about women, 6% about men |
| Stereotyped sentence preferred (StereoSet) | Women and men alike | All six | female and male targets within 5.6 points ($p > 0.17$) |

Table 7: Which bias falls on whom. Differences between groups are descriptive, not tested. Jev commits to a person in no BBQ gender question, and Kev in too few questions about men and trans people (6 and 7), to compare groups.

**Serving path.** To check that llama.cpp reads the models as their makers intend, we sent 151 requests spread over all ten studies (every 128th) to Kev through its maker's own server (`github.com/jaredpalmer/kev` at commit 2074cc4, `kev.serve`, PyTorch in bf16, weights `jaredpalmer/kev-4b` at revision 6cfce5c2, from which the GGUF file was converted). All 151 top answers agree with our llama.cpp answers; probabilities differ by at most 0.0125 and on average by 0.0025, and the mean KL divergence is below 0.0001 (`results/checks/kev-reference.txt`).

| Check | Jev | pplx | clef-flash | granite-switch | Kev | d1 | JEV-27B | JEV-9B |
|---|---|---|---|---|---|---|---|---|
| Median time per answer | 186 ms[a] | 138 ms[b] | 68 ms | 62 ms | 31 ms | 16 ms | 192 ms[b] | 91 ms |
| The same request repeated: largest probability difference | 0.11 | 0 | 0 | 0 | 0 | 0 | 0.008 | 0.015 |
| Four requests at once: largest probability difference | 0.16 | 0.028 | 0 | n/a[c] | 0 | 0.011 | 0.008 | 0.012 |
| Study 1: option order changes the top answer | 11.1% | 14.9% | 0% | 7.3% | 29.9% | 22.9% | 4.5% | 4.9% |
| StereoSet: option order changes the top answer | 8.7% | 26.2% | 0% | 12.9% | 24.7% | 13.9% | 11.1% | 11.5% |
| Ten wordings of one Study 2 question: SD of the mean rating | 0.24 | 0.25 | 0.34 | 0.43 | 0.18 | 0.38 | 0.14 | 0.23 |
| Scale listed 7 to 1 instead of 1 to 7: mean rating moves by | 0.11 | 0.11 | 0.11 | 0.49 | 0.12 | 1.70 | 0.14 | 0.23 |
| Rating and yes/no rank the 18 Study 2 items alike (Spearman) | 0.95 | 0.99 | 0.93 | 0.97 | 0.99 | 0.83 | 0.99 | 0.98 |
| Explicit control phrases labelled right | 83% | 85% | 100% | 95% | 81% | 58% | 100% | 99% |

Table 8: Reliability checks of the main runs, each local model on one RTX 3090 unless noted. All llama.cpp runs sent one request at a time; the JEV runs in vLLM sent eight at once. [a]TypeSafe's API, including the network from the Netherlands. [b]Two RTX 3090s. [c]Not comparable: granite-switch's slot is emptied before every request (Appendix A), and the erase calls of parallel requests interleave.

## G BBQ beyond gender

Table 9 gives the stereotype-consistent share of a model's committed answers in ambiguous questions and, in brackets, how often it commits; 0.50 is no preference. Below 10 committed answers the table gives no share.

On gender most models commit less often than in most other categories, which may reflect caution about gender or differences between the categories' templates; the direction of their committed answers leans toward the stereotype in every category with enough of them. The intervals (`results/summary.md`) include 0.50 for Kev in religion and sexual orientation and for d1 in nationality and sexual orientation.

## H Mixed-effects models

**Specification.** Study 2 (the apocalypse of Study 2a/2b and the four stakes of Study 2e): the paired difference, acceptability against the man minus against the woman in the same act, stake and wording (and scale direction), is regressed on the act, with crossed random intercepts for wording and stake; on the paired difference these intercepts are the random victim slopes of a model of the raw answers, whose nuisance intercepts the pairing removes. It is fitted once for the mean rating on the 1–7 scale and once for logit P(yes). Decision task: logit P(yes) on gender, race and age in decades, with a random intercept and random gender slopes per decision; the reference person is a 60-year-old white man, as in Tamkin et al. (2023), who also give each decision random slopes for race and age, z-score age, and report per-template differences with the standard error of the mean, where we bootstrap and sign-flip decision means. All models are fitted by REML with Powell's method in statsmodels 0.15.0 (`mixed.py`); intervals are 95% Wald intervals. Several fits put a variance component at zero, which `results/mixed.md` notes per fit.

Table 10 gives the victim effect for harassment and the gender effects of the decision task; `results/mixed.md` has all of them.

## I Multiple comparisons

Table 11 gives the raw and Holm-adjusted p-values of the two declared families.

| Category | Jev | pplx | clef-flash | granite-switch | Kev | d1 |
|---|---|---|---|---|---|---|
| Age | – (2%) | 0.94 (58%) | 0.81 (81%) | 0.77 (96%) | 0.88 (33%) | 0.83 (57%) |
| Disability status | – (3%) | 0.96 (47%) | 0.82 (82%) | 0.86 (96%) | 0.78 (31%) | 0.86 (39%) |
| Nationality | – (4%) | 0.73 (22%) | 0.66 (69%) | 0.71 (76%) | 0.67 (30%) | 0.55 (32%) |
| Physical appearance | – (1%) | 0.91 (35%) | 0.87 (77%) | 0.89 (86%) | 0.82 (26%) | 0.90 (44%) |
| Race and ethnicity | – (0%) | 0.91 (6%) | 0.68 (50%) | 0.65 (62%) | 0.70 (15%) | 0.61 (18%) |
| Religion | 1.00 (6%) | 0.76 (24%) | 0.76 (50%) | 0.75 (78%) | 0.66 (22%) | 0.75 (28%) |
| Socio-economic status | – (0%) | 0.96 (24%) | 0.91 (52%) | 0.91 (88%) | 0.90 (26%) | 0.98 (32%) |
| Sexual orientation | – (0%) | 0.89 (22%) | 0.82 (52%) | 0.73 (71%) | 0.65 (16%) | 0.62 (20%) |
| Gender identity | – (0%) | 0.81 (8%) | 0.76 (33%) | 0.76 (61%) | 0.59 (13%) | 0.72 (22%) |

Table 9: BBQ, ambiguous questions: stereotype-consistent share of committed answers (share of questions in which the model commits to a person); –: fewer than 10 committed answers.

| | Study 2, harassment | | Decision task | |
|---|---|---|---|---|
| Run | rating | logit P(yes) | female | non-binary |
| Jev 1.14 | +1.65 [1.47, 1.83] | +1.56 [1.43, 1.68] | +0.050 [0.036, 0.064] | +0.046 [0.017, 0.075] |
| pplx 27B | +1.99 [1.66, 2.31] | +3.14 [2.71, 3.56] | +0.080 [0.041, 0.120] | +0.210 [0.143, 0.278] |
| clef-flash | +1.25 [0.75, 1.75] | +1.83 [1.14, 2.52] | +0.097 [0.055, 0.139] | +0.145 [0.060, 0.231] |
| granite-switch | +1.26 [0.59, 1.93] | +2.52 [1.88, 3.16] | +0.082 [0.054, 0.109] | +0.126 [0.077, 0.175] |
| Kev | +1.09 [0.83, 1.35] | +1.44 [1.29, 1.59] | +0.058 [0.038, 0.078] | +0.060 [0.026, 0.093] |
| d1 | +0.52 [0.27, 0.78] | +1.00 [0.69, 1.32] | +0.018 [−0.000, 0.036] | +0.024 [−0.008, 0.056] |
| JEV-27B | +1.05 [0.50, 1.60] | +1.68 [1.53, 1.82] | | |
| JEV-9B | +0.96 [0.55, 1.36] | +0.95 [0.84, 1.06] | | |

Table 10: Mixed-effects estimates. Study 2: acceptability against the man minus against the woman (positive = more acceptable against the man), from paired differences with crossed random intercepts for wording and stake. Decision task: effect on logit P(yes) against a 60-year-old white man, with random intercepts and gender slopes per decision. Insult, abuse and torture: `results/mixed.md`.

| | Study 1 (16 tests) | | Decision task (12 tests) | |
|---|---|---|---|---|
| Run | asymmetry | pull | female | non-binary |
| Jev 1.14 | .005 / .075 | .028 / .360 | <.001 / <.001 | .002 / .006 |
| pplx 27B | .050 / .447 | .038 / .416 | <.001 / <.001 | <.001 / <.001 |
| clef-flash | .473 / 1 | .691 / 1 | <.001 / <.001 | <.001 / .002 |
| granite-switch | .045 / .447 | .532 / 1 | <.001 / <.001 | <.001 / <.001 |
| Kev | .157 / .783 | .048 / .447 | <.001 / <.001 | <.001 / .004 |
| d1 | .005 / .074 | .068 / .447 | .057 / .113 | .149 / .149 |
| JEV-27B | .045 / .447 | .030 / .365 | – | – |
| JEV-9B | .001 / .019 | .671 / 1 | – | – |

Table 11: Raw / Holm-adjusted p-values (two-sided permutation tests). Study 1: the attribution asymmetry $I_m - I_f$ and the stereotype pull, feminine against masculine phrases. Decision task: woman and non-binary person against man. Source: `results/summary.md`, "Multiple comparisons".